

A NEW MODEL OF ENERGY HUBS AND MULTI-OBJECTIVE OPTIMISATION FOR FUTURE NET-ZERO DISTRIBUTION NETWORKS

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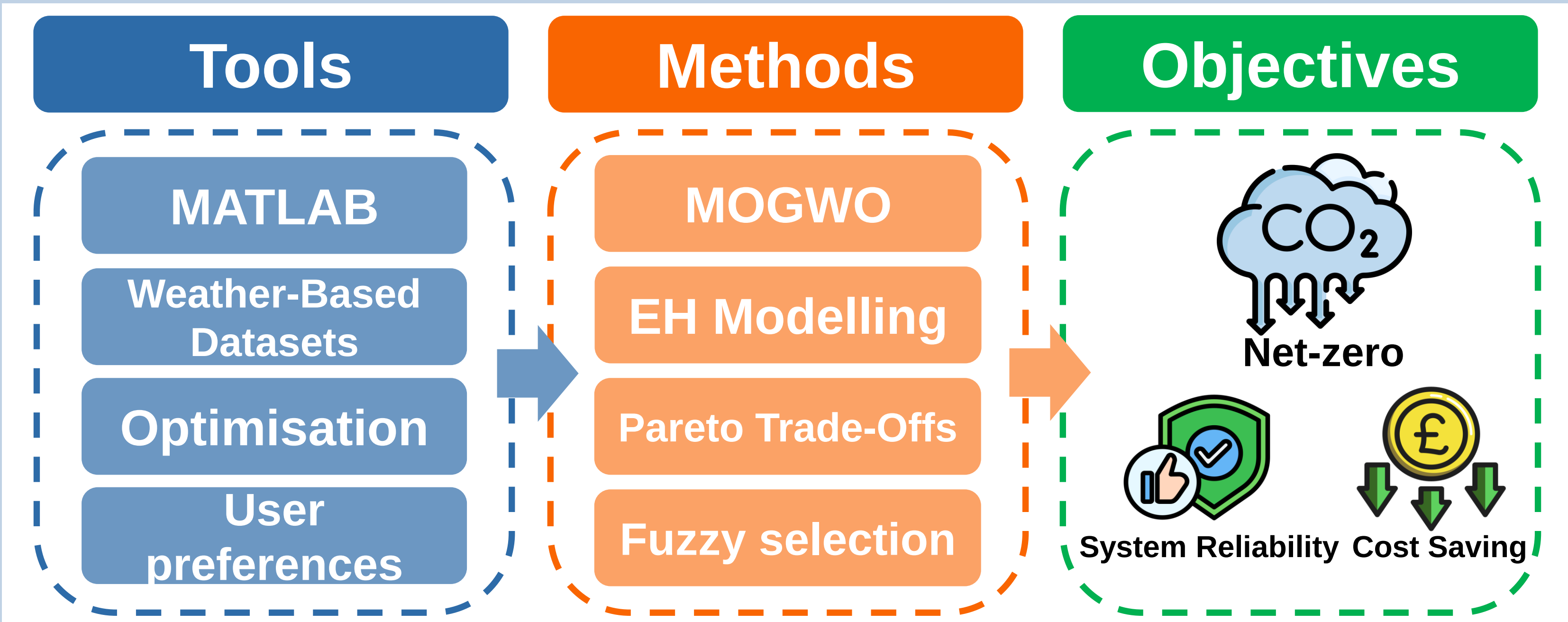
Introduction

The UN Environment Programme (UNEP) stresses the urgent need for energy sector transformation to achieve net-zero emissions. Energy production accounts for 71% of global emissions, yet siloed approaches have led to high costs and slow decarbonisation.

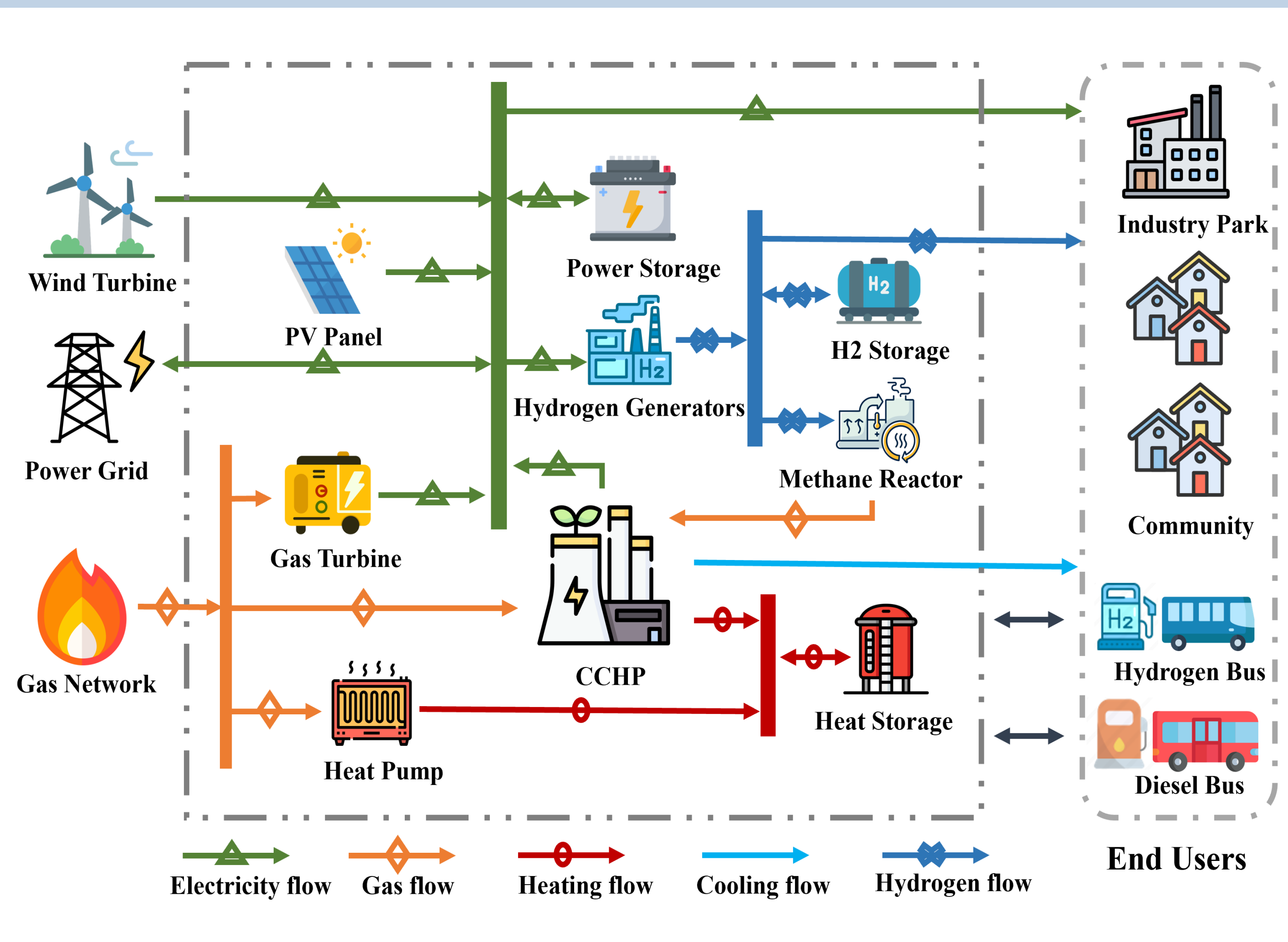
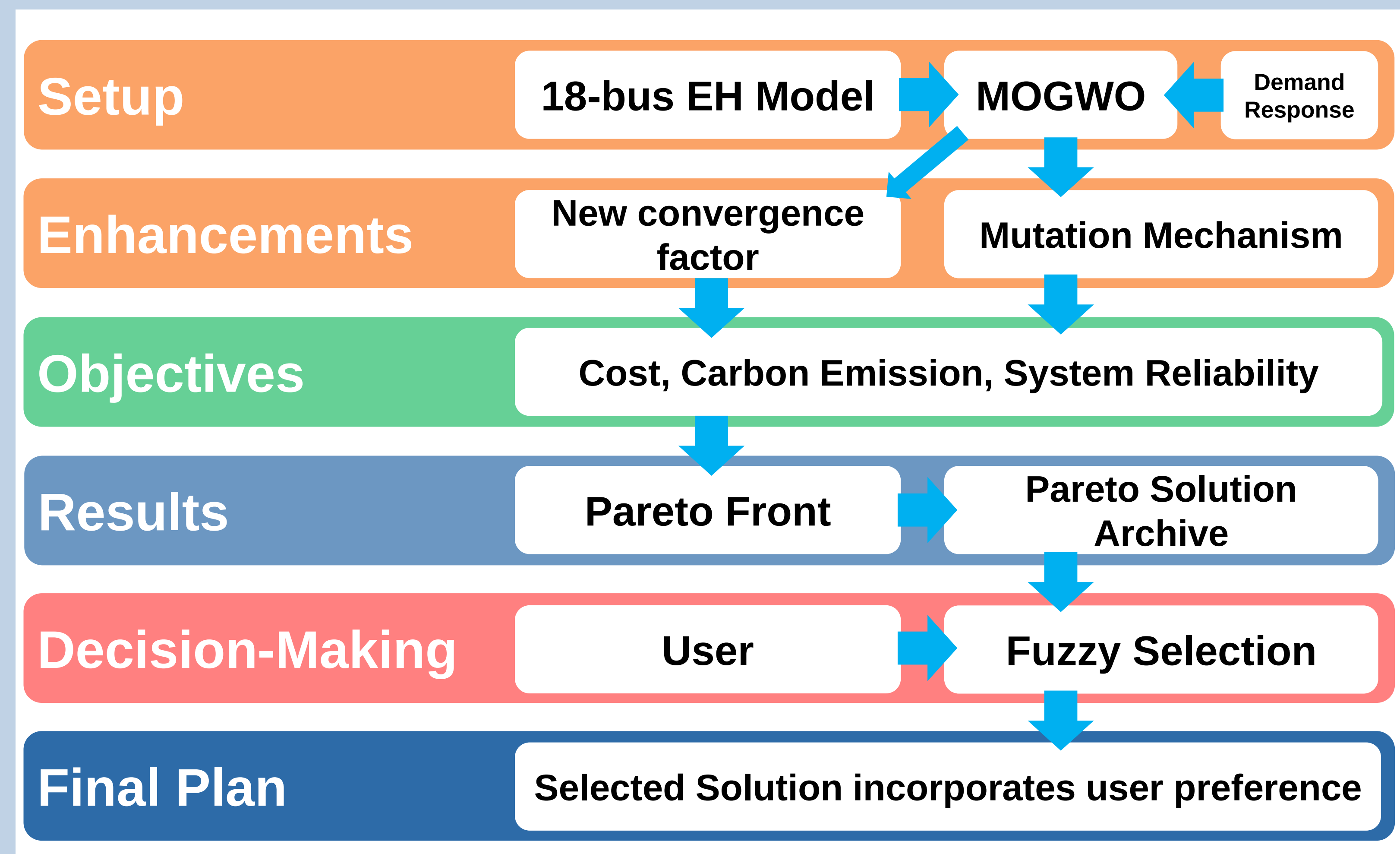
To address these challenges, **Multi-Energy Systems (MESS)** offer a coordinated approach by integrating electricity, heat, cooling, and fuels, enhancing efficiency and sustainability. The **Energy Hub (EH)** model optimises MESS through effective energy conversion, but existing traditional multi-objective optimisation algorithm struggle with local optima and poor solution diversity.

Objectives

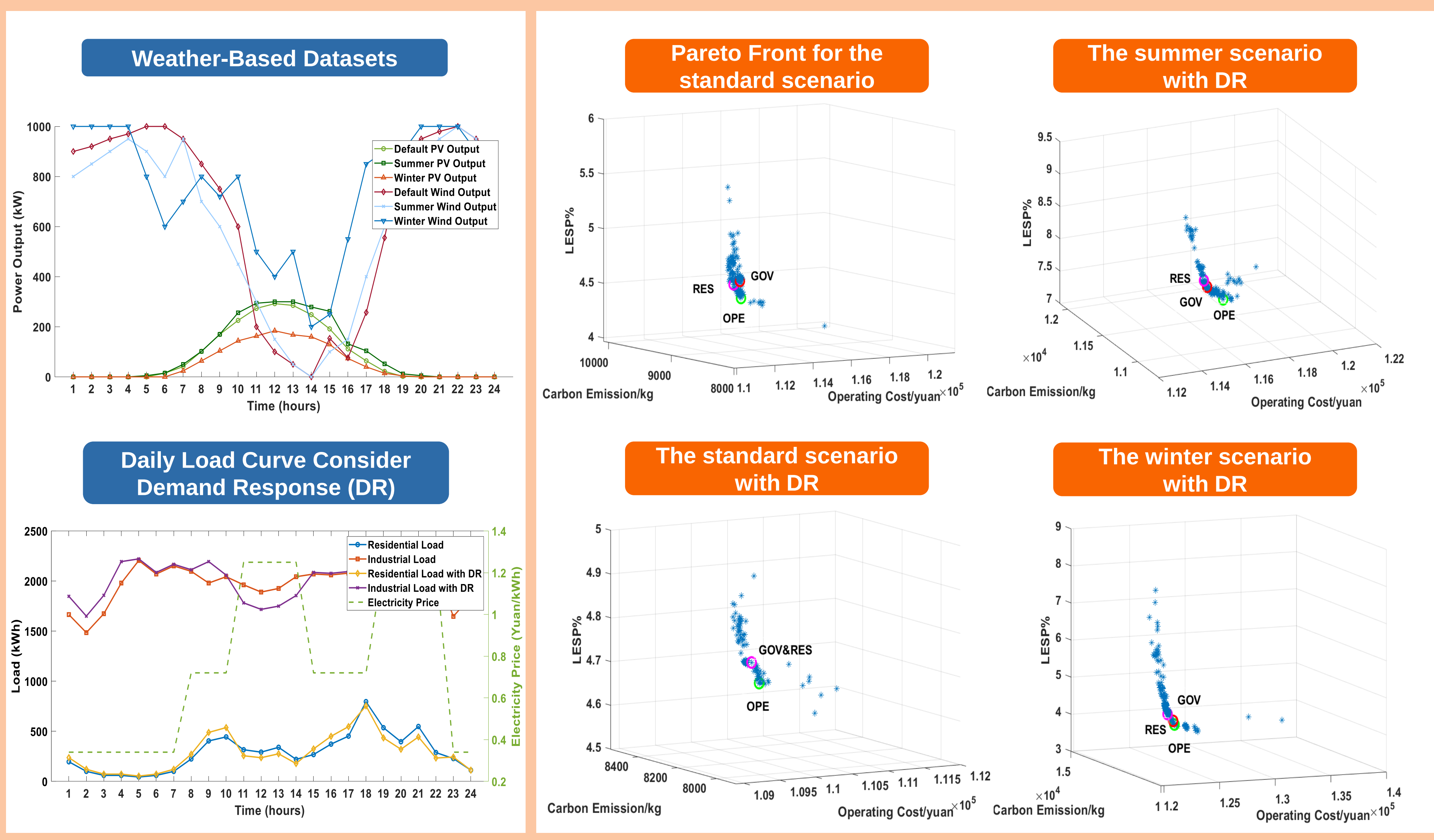
This study proposes an **enhanced Multi-Objective Grey Wolf Optimiser (MOGWO)** to optimise an eco-industrial park and residential community, achieving a cost-effective, reliable, and low-emission energy system. The proposed approach supports an efficient transition towards net-zero.



Methods



Results



Conclusion

Simulation-based case studies confirm the method's adaptability to real-world applications, offering practical insights for industrial use. The results show its potential as a valuable tool for Multi-Energy Systems planning, supporting the development of future net-zero energy solutions.

Future Work

- Extend the model to larger-scale multi-energy networks, considering coordination and energy transfer between EHs.
- Integrate stochastic modelling to account for renewable energy variability and demand fluctuations, improving system adaptability.